

1. Let $\phi : [0, 1] \rightarrow [0, 1]$ be a continuously differentiable increasing function with $\phi(0) = 0, \phi(1) = 1$, $F : [0, 1] \rightarrow [0, 1]$ an absolutely continuous function, $K = \sup_{c \in [0, 1]} \phi'(c)$, and $G = \phi \circ F \circ \phi^{-1}$. We claim that $G \in BV[0, 1]$. Note that for an arbitrary partition $0 = x_1 < x_2 < \dots < x_{n-1} < x_n = 1$ of $[0, 1]$ we have that $\{y_i\} = \{\phi^{-1}(x_i)\}$ is also a partition of $[0, 1]$ by virtue of ϕ^{-1} being an increasing function preserving the end points. Therefore $\sum_{i=1}^{n-1} |G(x_{i+1}) - G(x_i)| \leq K \sum_{i=1}^{n-1} |F(y_{i+1}) - F(y_i)| \leq K \|F\|_{BV}$, which implies that $\|G\|_{BV} \leq K \|F\|_{BV}$, giving us that $G \in BV[0, 1]$. Now to show that our choice of K is the optimal one, note that it must hold for all absolutely continuous functions on the unit square. Therefore it must hold for $F = x$. If we take $\alpha < K$ to satisfy the inequality $\|G\|_{BV} \leq \alpha \|F\|_{BV}$, then we have for all possible partitions $\{x_i\}_{i \in [n]}$ that $\sum_{i=1}^{n-1} |\phi(F(x_{i+1})) - \phi(F(x_i))| \leq \alpha \sum_{j=1}^{m-1} |F(y_{j+1}) - F(y_j)|$, where $\{y_j\}_{j \in [m]}$ is the optimal partition. Note that we can refine the partition $\{x_i\}$ with an infinite number of pairs of points which correspond to the slope K . Note that we can also refine the optimal partition by taking those same pairs and inserting them into $\{y_j\}$. Note that at some point there's going to be a critical mass where $|\phi(F(z)) - \phi(F(w))| = K|F(z) - F(w)| < \alpha|F(z) - F(w)|$, which we can do due to the mean value theorem which is a contradiction. Thus K is optimal.

2. Let F, G be absolutely continuous on $[0, 1]$. We want to show that FG is absolutely continuous. Let $\epsilon > 0$, then there exists $\delta_{1\epsilon}, \delta_{2\epsilon} > 0$ and let $\delta = \min\{\delta_{1\epsilon}, \delta_{2\epsilon}\}$ such that if $\{(a_j, b_j)\}_{j \in [n]}$ is a collection of disjoint intervals contained in $[0, 1]$, where $\sum_{j=1}^n |b_j - a_j| < \delta$, then $\sum_{j=1}^n |F(a_j) - F(b_j)|, \sum_{j=1}^n |G(a_j) - G(b_j)| < \epsilon$. Note that since F, G are absolutely continuous then they are uniformly continuous. Since they are defined over a finite interval, then there exists $M_F, M_G > 0$ such that $|F| \leq M_F, |G| \leq M_G$. Note that

$$\begin{aligned} \sum_{j=1}^n |F(a_j)G(a_j) - F(b_j)G(b_j)| &= \sum_{j=1}^n |F(a_j)G(a_j) - F(a_j)G(b_j) + F(a_j)G(b_j) - F(b_j)G(b_j)| \\ &\leq \sum_{j=1}^n |F(a_j)||G(a_j) - G(b_j)| + |G(b_j)||F(a_j) - F(b_j)| \\ &\leq \sum_{j=1}^n M_F |G(a_j) - G(b_j)| + M_G |F(a_j) - F(b_j)| \\ &< (M_F + M_G)\epsilon \end{aligned}$$

Therefore FG is absolutely continuous.

3. Let μ be the lebesgue measure on $\mathbb{R}$, $T : [0, 1] \rightarrow \mathbb{R}$ be an absolutely continuous function, A a borel set in $[0, 1]$ such that $\mu(A) = 0$. To show that $\mu(T(A)) = 0$, we will show for arbitrary $\epsilon > 0$ that we can find a cover via disjoint open intervals $\{a_j, b_j\}_{j \in [n]}$ whose length is small enough such that $\mu(T(A)) \leq \sum_{j=1}^n \mu(T((a_j, b_j))) \leq \epsilon$. Let $\epsilon > 0$. Then we have that there exists $\delta_\epsilon > 0$ upon which the collective length of a collection of disjoint intervals being less than δ_ϵ implies the measure with respect to T is ϵ . Note that since E is Borel and bounded we can find exactly that collection of interval such

that $A \subset \bigcup_{j=1}^n (a_j, b_j)$, $\mu(\bigcup_{j=1}^n (a_j, b_j)) < \delta_\epsilon$. Therefore

$$\mu(T(A)) \leq \sum_{j=1}^n \mu_T((a_j, b_j)) = \sum_{j=1}^n |T(b_j) - T(a_j)| < \epsilon.$$

Since ϵ is arbitrary then $\mu(T(A)) = 0$.

4. We want to prove that $F : \mathbb{R} \rightarrow \mathbb{C}$ is Lipschitz if and only if F is absolutely continuous with a bounded derivative.

- ($\Rightarrow$) Suppose that F is Lipschitz. Then there exists $L > 0$ such that for all $x, y \in \mathbb{R}$ we have that $|F(x) - F(y)| < L|x - y|$. Let $\epsilon > 0$. Note that if we have a set of disjoint intervals $\{(a_j, b_j)\}_{j \in [n]}$ with $\sum_{j=1}^n |a_j - b_j| < \frac{\epsilon}{L}$ then we have that $\sum_{j=1}^n |F(a_j) - F(b_j)| < L \sum_{j=1}^n |a_j - b_j| < L \frac{\epsilon}{L} = \epsilon$. Therefore F is absolutely continuous. Now suppose for contradiction that F has an unbounded derivative. Then there exists a sequence in which $\lim_{n \rightarrow \infty} |F'(x_n)| = \infty$. Note that there exists $N' \in \mathbb{N}$ such that for all $m \geq N'$, $|F'(x_m)| > L$. However, since F is Lipschitz we have that for all h , $|\frac{F(x_m+h) - F(x_m)}{h}| < L$, which by the definition of the derivative and the order limit theorem gives us $\lim_{h \rightarrow 0} |\frac{F(x_m+h) - F(x_m)}{h}| = |F'(x_m)| \leq L$. This is a contradiction since $|F'(x_m)| > L$. Thus F has a bounded derivative.
- ($\Leftarrow$) Suppose that F is absolutely continuous and has a bounded derivative. We want to show that it is Lipschitz. Let $x, y \in \mathbb{R}, x < y$, and $K = \sup |F'|$. If we consider only the real part of F , $G = \text{Re}(F)$, we have that $|G(x) - G(y)| = |\int_{[x,y]} G' dm| \leq Km([x, y]) = K|x - y|$. Note that the same can be done for the imaginary part, thus we have that $|F(x) - F(y)| \leq K\sqrt{2}|x - y|$, making F Lipschitz